

9 Handling Multi-Touch

The iPhone screen is a touch-sensitive display, which can detect multiple points of input and track them independently. It increases the iPhone's usability by allowing natural interaction between users and your applications. The applications can detect a wide range of gestures such as taps, swipes and pinches.

In this chapter, we will see how to detect touches in your application and then implement some of the other features that improve the interaction between the user and the application. We will also look at the underlying architecture that lets you detect gestures. The great part about UIKit is that it is very straightforward to add these types of interactions to our applications.

9.1 Detecting touches

Touch for the iPhone screen signifies a finger being placed on certain part of the screen. Multiple touches together make up a gesture, which is nothing but a sequence of events happening from the time you first touch the screen with one or more fingers and until none of them are in contact with the screen. An event containing information about the touch or touches that occurred is generated every time you interact with the multi-touch screen and your gestures are passed to the system inside it.

But, before we start detecting touches in our application, we need to acquaint ourselves with taps. A tap happens when the user touches the screen with a single finger and immediately lifts it up without moving it on the screen. Events handle the detection of touches in an iPhone and even keep track of the number of times a user has tapped, without any intervention from the developer as to the timing between multiple taps and differentiating between two single-taps and a double-tap. Lets dive into the architecture and learn how to detect a tap.

Create a View-based application project in Xcode. We are naming it TouchDetection. You can choose a different name, but make sure to account for it in the subsequent instructions. Add an image in the Resources folder by dragging and dropping. We will use it as the interaction point to detect a tap. The image we've used is Digit.jpeg. Next, open the TouchDetectionViewController.xib file in the Interface Builder by double clicking on it. Choose the UIImageView view in the View window and enlarge it to cover the entire View window. In the Attributes window, set the Image